Exp: 12. (a)

ECHO CLIENT - SERVER USING TCP SOCKETS.

AIM:

To implement an echo client - server application using TCP sockets.

ALGORITHM:

SERVER:

1. Create a TCP socket.
2. Bind the it to an IP address and port.
3. Listen for client connection.
4. Accept the client connection.
5. Receive the message and send it back
6. close the connection.

CLIENT:

1. Create a TCP socket.
2. Connect to the server.
3. Send a message.
4. Receive a echoed message.
5. Display the message
6. Close the connection.

Input:

Enter message: Hello Server

Output:

Server : Hello server.

CODE:

SERVER

```
import socket
S = socket.socket()
s.bind ("localhost", 5000)
s.listen()
c, addr = s.accept()
msg = c.recv(1024).decode()
c.send(msg.encode())
c.close()
s.close()
```

CLIENT

```
import socket
S = socket.socket()
s.connect(("localhost", 5000))
msg = input("Enter message:")
s.send(msg.encode())
print("Echo:", s.recv(1024).decode())
S.close()
```

RESULT:

Thus the echo-client-server was successfully implemented using the TCP sockets.